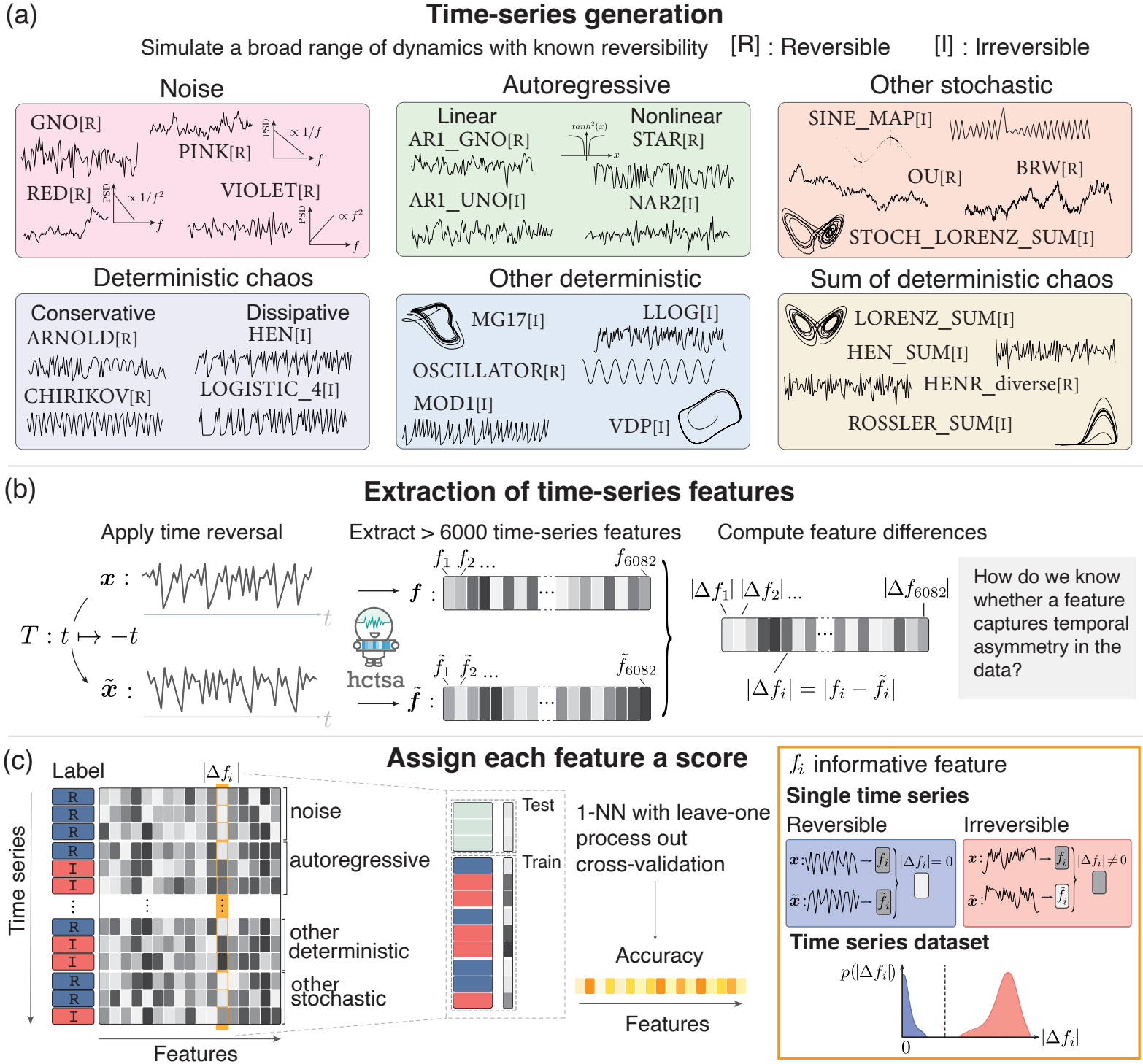
(a)
Time-series generation
Simulate a broad range of dynamics with known reversibility
[R] : Reversible
[I] : Irreversible
Noise
GNO[R]
PINK[R]
RED[R]
VIOLET[R]
Autoregressive
Linear
AR1_GNO[R]
AR1_UNO[I]
Nonlinear
STAR[R]
NAR2[I]
Other stochastic
SINE_MAP[I]
OU[R]
BRW[R]
STOCH_LORENZ_SUM[I]
Deterministic chaos
Conservative
ARNOLD[R]
CHIRIKOV[R]
Dissipative
HEN[I]
LOGISTIC_4[I]
Other deterministic
MG17[I]
LLOG[I]
OSCILLATOR[R]
MOD1[I]
VDP[I]
Sum of deterministic chaos
LORENZ_SUM[I]
HEN_SUM[I]
HENR_diverse[R]
ROSSLER_SUM[I]
(b)
Extraction of time-series features
Apply time reversal
Extract > 6000 time-series features
Compute feature differences
hctsa
How do we know whether a feature captures temporal asymmetry in the data?
(c)
Assign each feature a score
Label
Time series
Features
noise
autoregressive
other deterministic
other stochastic
Test
Train
1-NN with leave-one process out cross-validation
Accuracy
Features
informative feature
Single time series
Reversible
Irreversible
Time series dataset